Which axioms survive: an audit of the twenty axioms of the programme against the four

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Abstract

The earlier work of the Master Equation programme was stated on twenty axioms: twelve describing a twelve-dimensional geometry and its fields, and eight describing the mathematics by which results were obtained from it. The first two papers of this series derived that geometry and its field equation from four sentences about what is observed, together with a principle of minimality. We audit the twenty against those two papers and classify each as a theorem already proved, a target owed by a named later paper, an axiom, or not an axiom. Eight are theorems of the first two papers, with corrections where the earlier wording does not survive. Seven are targets, and for each the paper that owes it is named. Three are not axioms: a method, a statement about provability, and an interpretation. Two are one statement, and it is the only one that the four axioms do not reach: that what is observed is a shadow, which does not carry all the information of the object that casts it. We state it as a fifth axiom, as an existence statement without uniqueness. The series therefore rests on five sentences, one principle and one measured scale.

1 What is observed, and what this paper does

The purpose of this paper is to say how many axioms the series needs. The first paper, M1, began from four observations that need no instrument, took them as axioms, and derived a twelve-dimensional geometry from them. The second, M2, derived the field equation on that geometry from the same four. The programme these papers rebuild was not written that way. It was stated on twenty axioms, and this paper goes through the twenty one at a time and asks of each whether it is still needed.

1.1 How the twenty arose

The twenty were not chosen at the outset. In the earlier work more than ninety quantitative statements were obtained, across particle physics, cosmology and number theory, and certain patterns kept recurring in the derivations: a fold with four corners, a drive that stretches and squeezes, a field and its mirror image, a singular object that became regular when seen in more dimensions. Each recurring pattern was written down as an axiom, and the list grew to twenty. A list assembled in that way records what was found useful; it does not show which of its entries are independent. The count of statements obtained is given here as history and is not a result of this paper.

The first two papers of the series change the position. If the geometry is derived, every axiom that describes the geometry is a theorem or it is false; if the field equation is derived, every axiom that describes the fields is a theorem or it is false. What remains after that subtraction is the set of statements the four axioms do not reach, and the purpose of the audit is to find that set.

1.2 The four axioms

We restate the four as M1 did.

Axiom 1 (closure)**.** The universe closes on itself, so that it can be aware of itself.

Axiom 2 (growth and decay)**.** The universe grows and decays as a function of itself.

Axiom 3 (order)**.** The order of events matters, so that each instance is unique and cannot be foretold.

Axiom 4 (drive)**.** The universe is in constant change; it never stands still.

One remark connects the four, and it is a theorem of M2 rather than a decoration. The operation that exchanges growth with decay is a reversal of the drive together with a quarter-turn of the internal fibre (M2 Proposition 2.6). A quarter-turn done twice is a half-turn, and in the complex coordinate of the plane a half-turn is multiplication by $e^{i\pi} = -1$. So two exchanges of growth and decay are one fold of the fibre, and the fold is what makes the pillowcase of M1: the exponential of the second axiom, the turn of the fourth and the angle π of the corners meet in the one relation $S^2 = -I$ (certificate `L4_S_sq_fold` of M2; re-derived as `L3_exchange_squared`). [R]

1.3 The result in plain words

Of the twenty, eight are already proved in M1 and M2. Seven are statements about spectra, partition functions and the passage between dimensions that later papers of the series must prove, and we name the paper that owes each. Three are not axioms at all. The remaining two are the same statement written twice, once in words and once in mathematics, and it is the one statement the four axioms do not supply.

That statement can be put as plainly as the four. We can see the shadow, but the shadow is missing information about what it is a shadow of. Two shadows can be identical and be cast by different objects; one object can cast shadows that look nothing alike. The four axioms say what the object is. They do not say that everything we observe, including what looks broken or singular, is a shadow of that object. That is the fifth axiom.

1.4 What this paper proves, and how it is organised

The results are these. The twenty axioms fall into four classes (Definition 2.1), and the class of each is given in Theorem 2.2 with its reason. The fifth axiom is stated in words (Axiom 5) and formally (Definition 5.2), as the existence of a regular twelve-dimensional configuration above every observed one; that the passage down loses information is a remark with its proof (Remark 5.3), and uniqueness is not asserted. The smallest case, the corner of the pillowcase, is worked by hand (Proposition 5.4). The minimal set of the series is stated in one sentence (Theorem 7.1).

The paper is organised as follows. Section 2 recalls what M1 and M2 established, in the form the audit uses, and defines the classes. Section 3 treats the axioms that are theorems, Section 4 those that are targets, Section 5 the fifth axiom, and Section 6 those that are not axioms. Section 7 states the minimal set and the list of targets. Appendix A reproduces the twenty axioms as they were stated; Appendix B is the cumulative table of symbols; Appendix C contains the certificates. As in M1 and M2, this introduction is intended to be followed by a first-year student; from Section 2 the register is that of a formal paper. Rigour is tagged [R] (proved, or a cited classical theorem), [D] (derived under a stated reading or under minimality), [M] (motivated) and [G] (open), as in M1.

The deflationary reading is stated here, where it belongs. This paper proves nothing new about the geometry or the field equation. It is bookkeeping: it shows which of the earlier axioms

are consequences of results already proved, and it identifies what is left. A verdict of "theorem" below is only as strong as the theorem of M1 or M2 it cites, with the readings and the minimal choices that theorem carries. A verdict of "target" is a statement that something has not yet been proved in the series.

2 The setting: what M1 and M2 established, and the classes

2.1 The results the audit uses

M1 formalises the four axioms as conditions on a compact Riemannian geometry $(T \times \Sigma) \times F$ under four numbered readings (M1 Definitions 2.1, 2.2(i), 2.5, 2.6), defines minimality by four criteria applied in order (M1 Definition 3.2), and proves that the minimal geometry is unique (M1 Theorem 8.1):

$$\mathcal{M}_{12} = (S^1_{\mathrm{Sol}} \times S^3) \times K_8, \qquad K_8 = \left[(\mathbb{CP}^2 \times S^2) \times_w (T^2/\mathbb{Z}_2)\right]^{\mathrm{Spin}^c}, \tag{1}$$

with monodromy $M_0 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, determinant line $\mathcal{O}(3)$, $\chi(K_8) = 12$, dimension twelve and no continuous modulus, the overall scale being a declared input. The steps used below are: the drive is unique at the least entropy, with eigenvalues $\varphi^{\pm 2}$ and entropy $2 \log \varphi$ (M1 Theorem 4.2), and fixes one corner while cycling the other three (M1 Proposition 4.4); the fibre algebra is the quaternions (M1 Proposition 5.2); the fibre is a torus folded by $z \mapsto -z$ to the pillowcase, a sphere with four cone points of angle π (M1 Proposition 5.5, Theorem 5.7), carrying $\mathbb{R}$, $\mathbb{C}_i$, $\mathbb{C}_j$, $\mathbb{C}_k$ at the corners and $\mathbb{H}$ elsewhere (M1 Proposition 5.6); the dimension is twelve (M1 Theorem 6.1); the base is $\mathbb{CP}^2 \times S^2$ with a Spin^c structure forced by $w_2(\mathbb{CP}^2) \neq 0$ (M1 Propositions 7.1 to 7.3) and corner lift signs $(-1)^{mn}$ (M1 Corollary 7.4); the spatial factor is S^3 (M1 Proposition 7.5); the aspect ratio of the fibre is $\tau_2 = 1.8466$ (M1 Proposition 8.2, whose stationarity conditions are hypotheses owed by a later paper).

M2 adds a fifth reading, that the metric is positive-definite and time is the orbit parameter of the drive (M2 Definition 2.1), and states the standing principle that wherever the axioms leave a choice the minimal option is taken and named. It proves that the two sheets of the growth law are one field ψ, a section of the Clifford module $M \cong \mathbb{H}^{32}$, with shadow $\bar{\psi} = \hat{\tau}\psi$ (M2 Definition 2.2); that $\omega = \psi + \bar{\psi}$ and $\Phi_{\mathrm{QC}} = \omega|\psi - \bar{\psi}|^2$ is the unique lowest-order even scalar vanishing on both degenerate loci (M2 Definition 2.3, Proposition 2.4); that the exchange τ is the quarter-turn flow reversal with τ^2 the fold (M2 Proposition 2.6) and that $\bar{\psi}$ has the opposite handedness to ψ (M2 Theorem 2.7); that the field equation is

$$G_{MN} = 8\pi G_{\mathrm{eff}}\, T_{MN} + \zeta\, T^{\Phi_{\mathrm{QC}}}_{MN} + \Lambda_{\mathrm{eff}}\, g_{MN} \tag{2}$$

(M2 Theorem 3.1); that $\zeta = \chi(K_8)\pi = 12\pi$ (M2 Theorem 4.1, Corollary 4.2); and that ω vanishes at the locked phase and averages to zero at the uncorrelated phase (M2 Propositions 5.1, 5.3).

2.2 The twenty axioms

The earlier statement has two tiers. The first has twelve entries, Σ-0 to Σ-8 with Σ-P, Σ-S and Σ-T; Σ-2 has eight parts, a to h, one for each component of the geometry, and Σ-3 has a part 3a. The second tier has eight, 𝕄-1 to 𝕄-8. They are reproduced in Appendix A with their source lines; the count is certified in `L2_twenty`. Most entries consist of a headline statement followed by consequences for physics that the headline is said to produce. We call these consequences *riders*.

Definition 2.1 (the classes)**.** An axiom of the earlier statement is of class **T** if its headline is a theorem of M1 or M2, the theorem being named; of class **TT** if it is a theorem target, a statement not proved in M1 or M2 and owed by a named later paper of the series; of class **A**

Table 1: The classification of the twenty axioms.

Axiom	Class	Reason, and where it is argued
Σ-0 Universal Symmetry	T	M2 Proposition 2.6, Theorem 2.7; exact for the exponent, not for the periods (Section 3.4)
Σ-1 The Manifold	T	M1 Theorem 8.1 (Section 3.1)
Σ-2 The Fundamental Shapes	T	M1 Theorems 4.2, 5.7, 7.6, 8.1 and Propositions 7.1 to 7.5, part by part (Section 3.1)
Σ-3 The Pillowcase, with 3a	T	in part: M1 Theorem 5.7, Proposition 5.6; M2 Definition 2.2 (Section 3.2)
Σ-4 The Three Fields	T	M2 Definitions 2.2, 2.3, Proposition 2.4; one field, not three (Section 3.3)
Σ-5 The Shadow Principle	A	the fifth axiom (Section 5)
Σ-6 Shape Determinism	T	M1 Definition 3.2(1), Theorem 8.1; one declared scale (Section 3.5)
Σ-7 Dimensional Simplicity	N	an interpretation (Section 6)
Σ-8 The ω-Drive	TT	field-level zeros in M2; spectral statement owed by M4
Σ-P Dimensional Permeability	TT	owed by M8 and M11
Σ-S Shadow Influence	T	headline: M2 Definition 2.2, Theorem 2.7; channels owed by M5, M17
Σ-T Dual Transparency	TT	owed by M8
$\mathbb{M}$-1 Observables as spectral data	TT	owed by M4
$\mathbb{M}$-2 The Shadow Completion	A	the same statement as Σ-5
$\mathbb{M}$-3 The proof template	N	a method
$\mathbb{M}$-4 The Gödel boundary	N	a statement about provability
$\mathbb{M}$-5 The partition function	TT	owed by M4 and M5
$\mathbb{M}$-6 The spectral data	TT	owed by M4
$\mathbb{M}$-7 The Sol monodromy	T	M1 Theorem 4.2, with one correction (Section 3.6)
$\mathbb{M}$-8 The Spin^c twist	TT	first clause M1 Proposition 7.3; the rest owed by M4, M5

if it is not reached by the four axioms and is declared; and of class **N** if it is not an axiom: a method, a statement about the theory, or an interpretation. Each axiom has exactly one class. The class is that of the headline; riders are listed with their own status. Where an entry is a single block with no headline, it is of class T only if every assertive clause of it is a theorem of M1 or M2.

The rule in the last sentence is the minimal one: it never credits an entry with more than has been proved. [D] (minimality)

Theorem 2.2 (the classification)**.** *Under Definition 2.1 the twenty axioms are classified as in Table 1: eight of class T, seven of class TT, two of class A which are one statement, and three of class N.*

Proof. The verdicts are argued one by one in Sections 3 to 6. The counts, and the fact that the classes partition the twenty, are `L2_class_counts` and `L2_partition`. [D] □

3 The axioms that are theorems

Each verdict in this section cites a theorem of M1 or M2 and inherits its tag. None is unconditional: each rests on the readings and the minimal choices of the theorem cited, and where the earlier wording says more than the theorem, the excess is named and moved to Section 4.

3.1 The manifold and its shapes: Σ-1 and Σ-2

Σ-1 states that the universe is twelve-dimensional with the geometry (1). This is M1 Theorem 8.1, at [D]. Two clauses of its wording are not part of the theorem. That the additional dimensions are "vast, not small" is a statement about the overall scale, which M1 does not derive and declares as its one input; it is neither a theorem nor a further axiom. That every singular phenomenon is regular in the full geometry is Σ-5, treated in Section 5.

Σ-2 states that the geometry consists of specific components, each necessary, and lists eight. The headline is M1 Theorem 8.1, whose proof names the axiom and the criterion that force each factor. The parts are as follows, each at the tag of the result cited. The plane $\mathbb{CP}^2$ is M1 Proposition 7.2, the least direct step of M1 and tagged so there. The sphere S^2 is M1 Proposition 7.1. The torus is M1 Proposition 5.5, and its aspect ratio is fixed with no modulus by M1 Proposition 8.2; the value is 1.8466, where the earlier statement has 1.86. The involution is M1 Proposition 5.5 and Theorem 5.7. The warped product is the second axiom itself under M1's second reading, and its exponent is M2 Theorem 4.1. The Spin^c structure is M1 Proposition 7.3. The time circle with its monodromy is M1 Theorem 4.2. The spatial sphere is M1 Proposition 7.5. The Euler characteristic $3 \cdot 2 \cdot 2 = 12$ is M1 Theorem 7.6.

One refinement of M2 must be carried into every later use of these axioms. The earlier statement speaks of "the $\mathbb{Z}_2$ involution" for two operations. The fold $z \mapsto -z$ makes the pillowcase. What separates the field from its shadow is the exchange τ, a quarter-turn with a reversal of the drive, whose square is the fold (M2 Proposition 2.6). On the four corners the fold fixes every corner, the drive fixes P_1 and cycles the other three, and the exchange fixes P_1 and P_4 and swaps P_2 with P_3 (`L3_fold_fixes_corners`, `L3_drive_corners`, `L3_exchange_corners`).

The riders of Σ-2 identify the components with physics: the isometries of $\mathbb{CP}^2$ with the strong force, three generations, the weak mixing angle, the Higgs mechanism, the hierarchy, and others. M1 names no gauge group and no particle, and uses Witten's theorem only as a cross-check. These identifications are targets and are listed in Section 4. One clause of the headline, that whether other geometries could produce other universes is undecidable from within, is a statement about the theory and is of the kind treated in Section 6; M1 itself does not claim that a non-minimal geometry fails the axioms.

3.2 The pillowcase: Σ-3 and 3a

Part 3a states that the fibre algebra is non-commutative over a commutative base and reduces to the real numbers at exactly one of the four corners. This is M1 Proposition 5.6 with Proposition 4.4, at [D]: the algebra is $\mathbb{R}$ at P_1, a complex line at each of the other three, and $\mathbb{H}$ at every point with trivial stabiliser.

The headline of Σ-3 states that the base has two walls and an axis of ω, making four sectors. Part of this is in hand: the pillowcase and its four corners (M1 Theorem 5.7), and two sheets ψ and $\bar{\psi}$ of opposite handedness (M2 Definition 2.2, Theorem 2.7). The rest is not. In M2, ω is a field, the even part of the sheet field, and not a coordinate axis; the exchange τ swaps the corners at the ψ and ω half-periods; and the word "wall" is used without a definition. The two-by-two table of sectors is therefore not a structure that M1 or M2 exhibits. We class Σ-3 as T for what is proved and record the re-expression of its wall picture in terms of τ as a target without an owner (Table 3). [D]

3.3 The fields: Σ-4

Σ-4 states that there are three fundamental fields, ω, ψ and $\bar{\psi}$, coupled through Φ_{QC}. The fields and the coupling are M2 Definitions 2.2 and 2.3 and Proposition 2.4, with two corrections that M2 settles. There is one field and not three: the shadow and ω are both determined by ψ, and the independent parts are its even and odd parts. And the square in the coupling is the modulus squared, $\Phi_{\mathrm{QC}} = \omega|\psi - \bar{\psi}|^2$, since ω is quaternion-valued.

One rider does not hold as written. The earlier statement asserts that Φ_{QC} is non-zero for a topological reason. In M2 the coupling vanishes at the real corner, wherever the sheets are locked (M2 Proposition 5.1), and wherever they coincide; read pointwise the rider is false. What M2 proves is weaker: a non-zero sheet field of definite handedness cannot equal its shadow (M2 Theorem 2.7). The rider needs a precise statement and a proof where it is first used, and is listed in Table 3.

3.4 Symmetry: Σ-0

Σ-0 states that the universe is exactly symmetric across the involution and that every asymmetry is an artefact of observation from one side. The exchange τ exists and is a symmetry of the Sol block (M2 Proposition 2.6), it is the involution of the second axiom ([D], a minimal choice of M2), and the two sectors it exchanges have opposite handedness (M2 Theorem 2.7), so that their handedness sums to zero (`L3_chirality_sum`).

The symmetry is exact for the warp exponent and not for the period lengths. M2 shows that the exponent 12π is the same for either sheet, and that the aspect ratio τ_2 is not a convention: the decay rate differs by τ_2 between the sheets. So the two sides differ in a real quantity, and "every asymmetry is an artefact" does not hold as stated. We class Σ-0 as T with this caveat, at [D]. Its riders, that total baryon number and total charge vanish across both sides, are not touched by M1 or M2 (Table 3).

3.5 No free parameters: Σ-6

Σ-6 states that everything is fixed by the shapes and that there are no free parameters. Its content is the first criterion of minimality, that a minimal geometry has no modulus (M1 Definition 3.2), together with the theorem that the minimal geometry exists and has none (M1 Theorem 8.1, Proposition 8.2). It is therefore a theorem relative to the four axioms with minimality, at [D], and not a further axiom. Two qualifications are part of the verdict. There is one dimensionful input, the overall scale, which M1 declares (`L2_inputs`). And the stationarity conditions behind the aspect ratio are hypotheses of M1 owed by M7. The rider, that each named physical quantity is an invariant of the geometry, is a target paper by paper.

3.6 The drive: 𝕄-7

𝕄-7 states the monodromy, its eigenvalues, its Lyapunov exponent, the fixed point of its Möbius action, and that it is Anosov. All are M1 Theorem 4.2 and Proposition 4.1, at [R] given the second criterion of minimality, with one correction: the eigenvalues are φ^2 and φ^{-2}, so the exponent is $2\log\varphi$, as M1 states, and not $\log\varphi$ as the earlier axiom has it. The mixing property and the arrow of time are M2 Proposition 5.3 and Theorem 2.7.

3.7 The shadow sector: Σ-S

The headline of Σ-S states that the second sector is not in a different dimension but is the mirror image of the first within the same geometry. This is M2 Definition 2.2 and Theorem 2.7: the shadow is a section of the same module over the same $\mathcal{M}_{12}$, with the opposite handedness. Of its three channels of influence, the second, through ω, holds by the definition of ω; the first, through gravity, has its form in (2), where the shadow enters the curvature through $\zeta T^{\Phi_{\mathrm{QC}}}$, and its identification with dark matter is owed by M17; the third, through a completion of a modular object, is owed by M5.

Table 2: The targets and the papers that owe them.

Statement	Owed by	In hand from M1 and M2
$\mathbb{M}$-6: the spectrum and its degeneracies $d_{\mathrm{odd}}(n)$	M4	nothing
$\mathbb{M}$-5: the partition function and its modular type; its completion	M4, M5	nothing
$\mathbb{M}$-1: observables as internal spectral data	M4	nothing
$\mathbb{M}$-8: what the Spin^c twist does to the partition function	M4, M5	that the twist is forced (M1 Proposition 7.3)
Σ-8: ω as a functional of a spectrum	M4	the two zeros of ω and one hump (M2 Propositions 5.1, 5.3)
Σ-P: gauge fields confined; punctures	M8, M11	the metric and Φ_{QC} are fields on $\mathcal{M}_{12}$ (M2 Definition 2.5)
Σ-T: exactly two fields cross every boundary	M8	the two geometric sources in (2)
Σ-S: the shadow as dark matter; the completion	M17, M5	the headline (Section 3.7)
Riders of Σ-2: gauge groups, generations, mixing	M8, M9, M13	the shapes (Section 3.1)
Riders of Σ-2: hierarchy, vacuum energy	M10	$\zeta = 12\pi$ (M2 Corollary 4.2)
Riders of Σ-2: fine-structure constant; Higgs	M14, M15	nothing
Stationarity conditions behind τ_2	M7	the cubic (M1 Proposition 8.2)
Completeness of the geometry; the completeness axiom of the black-hole paper	M11	nothing stated here

4 The axioms that are targets

A target is a statement that M1 and M2 do not prove and that a later paper of the series owes. Table 2 names the owing paper for each of the seven axioms of class TT and for the riders of the axioms of class T. The assignment follows the order of the series, in which nothing is used before the paper that derives it.

Three remarks on the table. First, Σ-8 is one-sided as it was written: it describes low ω as ordered and high ω as interconnected and omits the second zero, at the uncorrelated phase, which M2 derives. Any later statement of it must carry both zeros.

Second, the function d_{odd} of $\mathbb{M}$-6 is the number of odd divisors of n, and not their sum as the earlier axiom says. The axiom itself fixes this, since it gives the Dirichlet series as $\zeta(s)^2(1-2^{-s})$, and

$$\sum_{n\geq 1}\frac{d_{\mathrm{odd}}(n)}{n^s} = \Big(\sum_{m\geq 1}\frac{1}{m^s}\Big)\Big(\sum_{d\ \mathrm{odd}}\frac{1}{d^s}\Big) = \zeta(s)\cdot\zeta(s)\big(1-2^{-s}\big)$$

when d_{odd} counts the odd divisors, each n being written as md with d odd in $d_{\mathrm{odd}}(n)$ ways. The smallest case by hand: 6 has the odd divisors 1 and 3, so $d_{\mathrm{odd}}(6) = 2$. [R]

Third, the completeness of the geometry is not stated in this paper. The question whether the completeness assumed in the treatment of the black-hole interior is a consequence of closure is left to the paper that treats it.

Table 3 lists five statements of the earlier axioms that no paper of the series is yet assigned to. They are targets, not premises.

5 The fifth axiom

5.1 What the four axioms do not reach

Σ-5 and $\mathbb{M}$-2 say the same thing: every apparent singularity in a lower-dimensional description is a projection artefact of a regular object on the full geometry, and the completed object is unique. Three assertions are bundled in it. The first is that what is observed is the image of

Table 3: Riders with no owing paper yet named in the series.

Statement	From
Total baryon number and total charge vanish across both sectors	Σ-0
The coupling Φ_{QC} is non-zero for a topological reason (precise form and proof)	Σ-4; used by 𝕄-2, 𝕄-3
The two homology generators of the torus as the origin of exactly two transparent fields	Σ-2, Σ-T
Punctures directed inwards	Σ-P
The wall picture and the four sectors re-expressed in terms of τ	Σ-3

something on $\mathcal{M}_{12}$ under a passage to fewer dimensions. The second is that this passage loses information. The third is that for everything observed, including what is singular, a regular object above it exists, and that it is unique.

The four axioms and M1 and M2 supply the first of these as a scheme, since M2 obtains four-dimensional statements by integrating over K_8 (M2 Definition 2.5). The second is a remark with a proof, given below. The third is not a consequence of the four axioms: they determine the geometry and the field equation, and they say nothing about which four-dimensional configurations arise. Its existence half is what the earlier work uses, and we declare it. Its uniqueness half we do not declare: it is not needed for the statement, the passage down is many-to-one, and in the series the object is fixed by the four axioms and minimality (M1 Theorem 8.1) and not by its shadow.

5.2 The axiom

Axiom 5 (shadow)**.** We can see the shadow, but the shadow is missing information about what it is a shadow of.

To use the sentence we must say what a shadow is, as M1 did for each of the four.

Definition 5.1 (configurations and the reduction)**.** A *configuration* is a set of the fields of M2 Definition 2.5 on $\mathcal{M}_{12}$ — the metric with its warp, the dilaton Φ_{QC}, the Spin^c spinor and the connection of the determinant line — satisfying the field equation (2). It is *regular* if every field is defined at every point of $\mathcal{M}_{12}$ and is smooth away from the corners and walls of the pillowcase, where M1 admits cone points and M2 admits a jump in the slope of the warp. The *reduction* ρ of a configuration is the four-dimensional data on $S^1 \times \Sigma$ obtained by integrating over K_8 in the two stages of M2 Definition 2.5. An *observed configuration* is four-dimensional data of the same kind, defined on $S^1 \times \Sigma$ away from a closed set of measure zero, on which it may fail to be defined or to be bounded.

Definition 5.2 (shadow, formal)**.** The geometry satisfies A5 if every observed configuration is the reduction of a regular configuration.

The definition is an existence statement. It does not say that the regular configuration is unique, and it does not name a mechanism by which it is found. It is the minimal option among those the earlier wording leaves open, and is recorded as such: the earlier axiom declared existence, uniqueness and two named mechanisms; we declare existence only. [D] (minimality)

Remark 5.3 (the reduction loses information). The reduction is not one-to-one on fields. Integration over K_8 sends to zero every function on K_8 of zero mean, so two sets of fields that differ by such a function in one component have the same four-dimensional data. [R] This is a statement about the map on fields. Whether two configurations, each satisfying (2), can have the same reduction is a question about the solutions of the field equation, which this paper does not answer; the axiom does not depend on the answer.

5.3 The smallest case by hand

The pillowcase is the smallest object of the series that has a singular point and a regular object above it, and every part of the axiom can be seen in it.

Proposition 5.4 (the corner of the pillowcase)**.** *Let $P = T^2/\mathbb{Z}_2$ be the pillowcase and $\pi : T^2 \to P$ the fold.*

(i) P has four cone points of angle π, at which all of its curvature is concentrated; each is the image under π of a point of the torus at which the torus is flat and regular.

(ii) The fold is two-to-one away from the four corners and identifies z with $-z$: it is not one-to-one.

(iii) The torus is not the only regular surface lying over P. For every $n \geq 2$ the torus $\mathbb{R}^2/n\mathbb{Z}^2$ maps onto $T^2 = \mathbb{R}^2/\mathbb{Z}^2$, and hence onto P, by a map that commutes with the fold.

Proof. (i) is M1 Theorem 5.7 and the paragraph that follows it there: near a fixed point the fold is the rotation by π, a disc of angle 2π is folded to a cone of angle π, and the torus is flat. (ii) The fold sends z and $-z$ to the same point, and $z \neq -z$ except at the four half-lattice points. (iii) The identity of $\mathbb{R}^2$ descends to a map $\mathbb{R}^2/n\mathbb{Z}^2 \to \mathbb{R}^2/\mathbb{Z}^2$, since $n\mathbb{Z}^2 \subset \mathbb{Z}^2$, and it commutes with $z \mapsto -z$; composing with π gives a map onto P. The finite content is certified on the grids $(\mathbb{Z}/4)^2$ and $(\mathbb{Z}/8)^2$: the fold is an involution with exactly four fixed points, the other twelve points fall into six pairs, two distinct points have the same image, every point of the smaller grid has four preimages in the larger, and the reduction commutes with the fold (`L1_fold_involution`, `L1_four_fixed`, `L1_two_to_one`, `L1_not_injective`, `L1_second_cover`, `L1_second_cover_fold`). [R] □

In the words of the axiom: the corner is singular in the pillowcase and regular in the torus (existence); the pillowcase does not record which of z and $-z$ a point came from (loss of information); and the pillowcase alone does not say which torus was folded (no uniqueness). The proposition is an illustration in dimension two and nothing later rests on it.

5.4 What justifies declaring it

The earlier work offered a justification, $\mathbb{M}$-4: that the statement cannot be proved within any single existing formal system, three approaches having each reached an obstruction. We keep the observation and limit what is claimed for it. That three approaches did not close is evidence that the statement is hard; it is not a proof that the statement is undecidable, and nothing in the earlier text exhibits it as independent of a named formal system. We therefore argue, and do not prove, that the statement is to be taken as an axiom, on the same ground as the four: it is an observation in plain words whose content the others do not supply. [M]

The axiom has a failure test, as every minimal choice of the series has. It fails if an observed configuration is shown to have no regular configuration above it. It is too weak if a later paper needs the uniqueness of the regular configuration, or a named mechanism of completion, as a premise and not as a result; in that case the choice made here is to be flagged and enlarged in the open. The first paper that tests it on a case beyond the pillowcase is M5.

6 What is not an axiom

Three of the twenty are of class N.

$\mathbb{M}$-3 states that each proof follows one template: identify the shadow, construct the completion, prove a lock, derive the consequence. This is a method. It is not a statement about the universe or about the mathematics, and a method is adopted, not assumed. Two of its four steps rest on statements that are not in hand: the completion, which is the fifth axiom, and the lock, which is the rider of Σ-4 listed in Table 3.

$\mathbb{M}$-4 states that the Shadow Principle cannot be proved from within. It is a statement about provability and not an axiom of the theory. Its use is confined to Section 5.4.

Σ-7 states that complexity is a projection artefact of simplicity. It has no formal statement, no result of the earlier work cites it as a premise, and neither the definition of minimality nor the theorem of the least dimension in M1 asserts it. It is an interpretation, close in spirit to the fifth axiom, and is recorded as such.

Three clauses of other axioms are of the same kind and are not counted separately: the clause of Σ-2 on the undecidability of other geometries; the clause of Σ-T that structured diversity requires two transparent fields, which is an argument; and the choice of unit in the earlier treatment of the warp, which M2 carries as a statement of units. No axiom is a convention as a whole; the conventions of the series are the assignment of two corner labels (M1 Appendix B), which sheet is called ψ, and the overall sign of ψ (M2).

7 The minimal set, and the list of targets

Theorem 7.1 (the minimal set)**.** *The series rests on the four axioms of M1, the fifth axiom of this paper, the principle of minimality, and one measured scale; of the twenty axioms of the earlier statement none other is independent.*

Proof. By Theorem 2.2, eight of the twenty are theorems of M1 and M2, which rest on the four axioms, the five readings and minimality; seven are targets, which are not premises; three are not axioms; and the remaining two are the one statement declared as A5 (`L2_one_axiom`, `L2_minimal_set`). The scale is the declared input of M1 Theorem 8.1 (`L2_inputs`). [D] □

The theorem is to be read with its limits. It says that no further axiom is needed to state what M1 and M2 have proved and what the later papers are to prove. It does not say that the targets of Table 2 will be reached from five axioms; if one of them proves not to be derivable it returns to the list of axioms, with the failure recorded. The five readings of M1 and M2 are definitions and not axioms, and a reader who takes one of them differently has a different theorem. And the count of "one measured scale" is of dimensionful inputs; the claim that there is no dimensionless one rests on the stationarity conditions of M1 Proposition 8.2, which M7 owes.

Tables 2 and 3 are the list of what remains. They are the content of the papers that follow: the spectrum and the partition function, the test of the fifth axiom on the pillowcase, the aspect ratio, the masses and mixings, the vacuum energy, the black-hole interior and the couplings.

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A The twenty axioms, as stated

The twenty axioms audited in this paper are reproduced here as data. The wording is that of the programme's earlier statement of them; that document is superseded by the present series and is not cited. Line numbers refer to its tenth version. An omission is marked [...]; words in square brackets replace a superseded name of ω. A paragraph headed *Note added in M3* is commentary of the present paper and is not part of the earlier wording. The twelve creation axioms come first (Σ-3a is a part of Σ-3), then the eight mathematical axioms.

The creation axioms

Σ-0: Universal Symmetry

Source lines 335–345.

The universe is exactly symmetric across the $\mathbb{Z}_2$ involution.

Every apparent asymmetry—matter/antimatter, L/D chirality, baryon excess, visible/dark—is a confinement artefact of observation from one wall of the pillowcase. The totality is perfectly balanced. Total baryon number across both walls: zero. Total chirality across both walls: zero. Total charge across both walls: zero.

Plain English: The universe is perfectly balanced. Everything that looks lopsided—more matter than antimatter, left-handed amino acids, visible matter versus dark matter—looks that way only because we are seeing one side. The other side has the mirror image. Together, perfect symmetry.

Mathematical lineage. Universal Symmetry is the axiomatic statement of a century of physics-led reasoning about what symmetry demands. Dirac (1928, *The Quantum Theory of the Electron*) [2] derived the relativistic wave equation for the electron and, confronting its negative-energy solutions, reasoned that symmetry required a mirror particle — the positron, confirmed by experiment four years later. Lüders (1954) [11] and Pauli (1955) [12] independently proved the CPT theorem: every Lorentz-invariant local quantum field theory is invariant under the combined operation of charge conjugation, parity, and time reversal — the deepest symmetry statement in physics, asserting that apparent asymmetries under any one operation must be balanced by the others. Philip W. Anderson (1972, "More is Different") [1] established broken symmetry as observational fact: the ground state of a many-body system need not respect the symmetry of its underlying laws, and what the observer measures is the state, not the law. Σ-0 extends this lineage by identifying the missing wall: apparent asymmetries — matter/antimatter, chirality, baryon excess, visible/dark — are not consequences of broken laws but confinement artefacts of observation from one wall of the $\mathbb{Z}_2$ pillowcase. […] The axiom therefore names the principle; it is an identification, not an invention.

Σ-1: The Manifold

Source lines 348–360.

We are four-dimensional entities living in a twelve-dimensional universe. The additional dimensions are vast, not small.

What we observe in four dimensions is a projection, and projections lose information. Every phenomenon that appears singular, paradoxical, or fine-tuned in the projection is smooth and natural in the full geometry. This is not the string-theory compactification paradigm of dimensions curled into Planck-scale tubes. The dimensions are vast.

Note added in M3. A picture for the reader: the universe is like a set of nested Russian dolls. We are in the fourth doll, and we are contained within eight other, larger dolls, each larger than the one before.

Formally: $\mathcal{M}_{12} = (S^1_{\text{Sol}} \times S^3) \times K_8$, where $K_8 = [(\mathbb{CP}^2 \times S^2) \times_w (T^2/\mathbb{Z}_2)]^{\text{Spin}^c}$, with Euler characteristic $\chi(K_8) = 12$.

Plain English: We are four-dimensional beings in a twelve-dimensional universe. Those extra dimensions are not tiny curled-up strings—they are vast. We are seeing a shadow of the full reality, and the shadow loses information.

Mathematical lineage. The Manifold is the axiomatic statement of a 200-year tradition that treats the dimensionality and intrinsic geometry of space as empirical rather than a priori. Gauss (1827, *Disquisitiones generales circa superficies curvas*) [5] established the Theorema Egregium: curvature is intrinsic — geometry need not depend on embedding, and can be known from measurements within the manifold alone. Riemann (1854, *Über die Hypothesen welche*

der Geometrie zu Grunde liegen, his Göttingen Habilitationsvortrag) [14] generalised Gauss's intrinsic geometry from two-dimensional surfaces to n-dimensional manifolds, reframing dimension itself from a priori fact to empirical question. Kaluza (1921, "Zum Unitätsproblem der Physik") [10] executed the first physical realisation: Einstein's equations in five dimensions automatically produce gravitation and electromagnetism in four, unifying the two forces through a single geometric structure. Σ-1 extends the lineage: the twelve dimensions of $\mathcal{M}_{12}$ are vast and topologically isolated, a direct realisation of Riemann's empirical dimensionality. […] The axiom therefore names the principle; it is an identification, not an invention.

Σ-2: The Fundamental Shapes

Source lines 363–427.

The twelve-dimensional manifold consists of specific geometric components. Each is fundamental and irreducible. Given these shapes, everything follows.

There are no free parameters within this geometry. The Master Equation, the Standard Model, the fine structure constant, the cosmological constant, dark matter, consciousness, and the spectral structure of the primes are all consequences of these specific shapes and their interactions. Each component is independently necessary: remove any one and the framework collapses.

Whether alternative self-consistent geometries could produce other universes is undecidable—not merely from within the MEF as a formal system, but from within this universe. The shapes are not a description of reality that we could in principle examine from outside. They *are* reality. We are inside them. This is the **External Horizon** of the Gödel Boundary.

Note added in M3. By M1 (Theorem 8.1, the minimality theorem) this is the minimal topology that satisfies the four axioms. There may well be higher dimensions of which this universe is a subset. That is unknowable from where we are.

Plain English: The universe is built from exactly these shapes. Each is specific and each is necessary. Everything we observe—every force, every particle, every constant—drops out of these shapes and nothing else. Could different shapes have made a different universe? We cannot know. Not because we haven't worked it out yet, but because we are inside these shapes—they are our reality, not a model we look at from outside.

Σ-2a: $\mathbb{CP}^2$—The Colour Engine

Complex projective 2-space. Isometry group = SU(3): this is the strong force. Dolbeault cohomology $H^{0,2}(\mathbb{CP}^2, \mathcal{O}(-4))$ gives exactly three fermion generations. $w_2(\mathbb{CP}^2) \neq 0$ forces Spin^c (Σ-2f). $c_1(\mathcal{O}(3))^2 = 9$ feeds into the topological weight $W = 36$ that determines the fine structure constant.

Remove $\mathbb{CP}^2$ and you lose: the strong force, three generations, all quark physics, the fine structure constant.

Σ-2b: S^2—The Electroweak Sphere

The unique simply-connected 2-manifold with SU(2) isometry. $\chi(S^2) = 2$. Electroweak breaking gives $\sin^2\theta_W = 0.224$. Geometric tension versus Spin^c monopole flux produces the Higgs mechanism.

Remove S^2 and you lose: the weak force, electroweak symmetry breaking, the Weinberg angle, the Higgs mechanism.

Σ-2c: T^2—The Torus

Two independent homology generators are the topological origin of the two transparent fields (gravity and ω). Doubly-periodic structure enables modular forms. Aspect ratio $\tau_2 = L_\omega/L_\psi = 1.86$ is fixed by Spin^c flux stabilisation, not a free parameter.

Remove T^2 and you lose: modular forms, the partition function, the entire spectral structure, the ω-axis.

Σ-2d: $\mathbb{Z}_2$—The Involution

The order-2 group acting on T^2 by $z \mapsto -z$, creating the orbifold $T^2/\mathbb{Z}_2$ (the pillowcase). Four fixed points define the four sectors of reality (Σ-3). Q-wall/AQ-wall separation is forced. Generates mock modularity. Creates all apparent asymmetry.

Remove $\mathbb{Z}_2$ and you lose: the shadow sector, dark matter, chirality, mock modularity, the four-sector pillowcase structure.

Σ-2e: $\times_w$—The Warped Product

The fibration connecting $\mathbb{CP}^2 \times S^2$ to $T^2/\mathbb{Z}_2$ is warped, not direct. Creates the Planck–electroweak hierarchy. Enables $\Lambda_{\text{eff}} \approx 10^{-122}\, M_{\text{Pl}}^4$.

Remove $\times_w$ and you lose: the hierarchy between fundamental forces, the cosmological constant explanation, the entire scale structure of physics.

Σ-2f: Spin^c—The Twist

The Spin^c structure on K_8, forced by $\mathbb{CP}^2$. Enables fermions. Breaks Liu rigidity. Determinant line bundle $L = \mathcal{O}(3)$ gives $\mathrm{U}(1)_Y$. Simultaneously creates the disease (mock modularity) and provides the cure (Zwegers completion).

Remove Spin^c and you lose: all fermions, all dynamics, hypercharge. The universe cannot exist.

Plain English: The pillowcase has a spin that creates rigidity. That rigidity is broken through asymmetry—the Spin^c twist. Without it, nothing would happen. The universe would be frozen. The twist is what makes everything move, evolve, live.

Note added in M3. The same condition says what the next time slice is to be, so that each time slice is different. In the series this is the content of A3 (each instance is unique) and of A4 (the universe never stands still), with the twist as their carrier. Time can move forward; without this condition everything would be frozen.

Σ-2g: S^1_{Sol}—The Temporal Circle

The circle carrying the Sol monodromy $M = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{Z})$. Eigenvalues ϕ^2 and ϕ^{-2}. Anosov property generates mixing, chaos, the arrow of time. $D(\tilde{M}) = 5 = \mathrm{disc}(\mathbb{Q}(\sqrt{5}))$.

Remove S^1_{Sol} and you lose: the arrow of time, mixing, chaos, Anosov dynamics, Fibonacci growth, the cascade pump in turbulence, the conservation law $F(\frac{1}{2}) = 1$.

Plain English: The Sol monodromy is the clock of the universe. It ticks at the golden ratio. Every new growth—in time, in nature, in a spiral—is displaced by ϕ. That is why we see Fibonacci everywhere, we live in a fractal universe.

Σ-2h: S^3—The Spatial Sphere

The unique simply-connected closed 3-manifold with constant positive curvature (Perelman's theorem, resolving the Poincaré Conjecture). Closed, finite spatial topology. Discrete spectrum. CMB low-ℓ suppression.

Remove S^3 and you lose: spatial closure, discrete spectrum, CMB low-ℓ suppression, closed-universe predictions.

The Euler Characteristic

$\chi(K_8) = \chi(\mathbb{CP}^2) \times \chi(S^2) \times \chi(T^2/\mathbb{Z}_2) = 3 \times 2 \times 2 = \mathbf{12}$. This single topological integer propagates through the entire framework: fermion generations $= \chi/4 = 3$; coupling $\zeta = \pi\chi = 12\pi$; hierarchy $e^{12\pi}$; cosmological constant exponent $e^{-96\pi}$ where $96 = 8 \times 12$; BAO spectral bridge $e^{-\zeta/\dim(K_8)} = e^{-3\pi/2}$.

Σ-3: The Pillowcase

Source lines 430–449.

The control plane ($T^2/\mathbb{Z}_2$) has two walls and a drive axis, creating four sectors.

The Quantum Wall (ψ): the visible, measurable sector. The Anti-Quantum Wall ($\bar{\psi}$): the shadow sector—structurally essential, not directly observable from the Q-wall. The ω-axis: [the Coherent Field], from low (ordered, classical) to high (interconnected, quantum).

	Low ω (ordered)	**High ω (diverse)**
ψ **(Q-wall)**	Standard Model, visible matter, classical physics	Life, consciousness, creativity, negentropy
$\bar{\psi}$ **(AQ-wall)**	Dark matter—gravitationally present, electromagnetically invisible	Dark consciousness—the shadow at maximum coherence

Plain English: The pillowcase has four areas. On the quantum wall: ordinary matter at low coherence, and living conscious beings at high coherence. On the anti-quantum wall: dark matter at low coherence, and dark consciousness at high coherence.

Σ-3a: The Quaternionic Layer

Source lines 452–473.

The fibre algebra of K_8 is non-abelian over an abelian base, and the non-commutativity resolves to the classical real limit at exactly one of the four corners.

The pillowcase $T^2/\mathbb{Z}_2$ carries two structurally distinct algebraic layers. The *base* layer is the Klein-four group $V_{\text{geom}} = (\mathbb{Z}/2)^2$ of half-period translations of T^2, whose four elements index the four corners; together with the centre $Z = \langle -1 \rangle$ of the quaternion group, the base structure $V_{\text{geom}} \times Z$ is abelian. The *fibre* layer is the quaternion algebra $\mathbb{H}$ acting on the pillowcase interior, with unit-quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ — the 8-point non-abelian multiplicative group. Non-abelian fibre, abelian base.

At each corner of the pillowcase the fibre algebra degenerates to a sub-algebra of $\mathbb{H}$. The four limits are

$$C_0 \longrightarrow \mathbb{R}, \quad C_i \longrightarrow \mathbb{C}_i, \quad C_j \longrightarrow \mathbb{C}_j, \quad C_k \longrightarrow \mathbb{C}_k,$$

with dimensions $1, 2, 2, 2$ respectively. Only at C_0 does the fibre reach the fully-commutative classical limit $\mathbb{R}$; the other three corners retain a non-trivial imaginary unit and remain complex. The four corners are therefore not symmetric under the $\mathrm{Aut}(\mathbb{H})$ action: C_0 is uniquely distinguished as the corner where the non-commutativity of the interior commutativises completely.

This asymmetry is the structural source of the three Millennium closures on $\mathcal{M}_{12}$. The Riemann critical line $\sigma = \frac{1}{2}$ is the fixed locus of the transpose anti-involution on the hyperfinite completion of the fibre algebra, and that fixed locus intersects $\mathbb{H}$ at exactly $\mathbb{R}$ — i.e., at C_0 […]. The Yang–Mills mass gap is the minimum geodesic distance on K_8 between distinct corners under the $\mathrm{Sp}(1)$-action on the fibre; the alternation $K_2(n, -1; 2) = (-1)^n$ is the orbit signature of

$V_{\mathrm{Aut}} \times Z$ on Q_8 over n topological wraps of the base [...]. The Navier–Stokes velocity field embeds into $\mathbb{H}$ as a pure-imaginary quaternion; the Zwegers completion $\hat{f} = f + g^*$ is the algebraic closure $\mathrm{Im}(\mathbb{H}) \hookrightarrow \mathbb{H}$ that re-admits the $\mathbb{R}$-component generated by Hamilton multiplication [...]. Each of the three closures uses a different structural feature of this single non-abelian-over-abelian algebra; the uniqueness of C_0 is what selects the Riemann critical line, the Yang–Mills vacuum corner, and the Navier–Stokes scalar-absorption channel.

Plain English: The pillowcase has a hidden algebraic structure that is deeper than the four sectors of Σ-3. The interior carries Hamilton's quaternions, where rotations do not commute — order matters. The base, which labels where you are, is commutative — order does not matter. At three of the four corners, the quaternion structure simplifies to ordinary complex numbers, keeping a trace of its quaternionic parent. At exactly one corner — the origin — it collapses all the way to the real numbers, fully commutative. That one unique corner is what makes the three Millennium problems solvable: it is where the Riemann critical line lives, where the Yang–Mills vacuum sits, and where the Navier–Stokes real-component absorption happens. The asymmetry of the four corners is not decorative. It is the load-bearing structural feature of the manifold.

Mathematical lineage. The Pillowcase is the axiomatic statement of a two-century tradition in non-commutative algebra and its classification. Hamilton (1843, Brougham Bridge) [6] constructed the quaternions $\mathbb{H}$ as the first non-commutative number system supporting geometric rotation — the only honest step beyond $\mathbb{C}$ for half a century. Jordan (1870, *Traité des substitutions et des équations algébriques*) [9] established the systematic theory of finite groups, whose classification methods and structural insights underlie all subsequent work on discrete symmetries — including the four-fixed-point structure of the pillowcase base. Frobenius (1877, *Über lineare Substitutionen und bilineare Formen*) [4] closed the division-algebra classification: $\mathbb{R}$, $\mathbb{C}$, $\mathbb{H}$ are the only finite-dimensional associative division algebras over $\mathbb{R}$; the octonions $\mathbb{O}$ extend the hierarchy non-associatively and do not feature in the pillowcase's algebraic spine. Σ-3 realises the associative sub-hierarchy $\mathbb{R} \subset \mathbb{C} \subset \mathbb{H}$ geometrically over the four corners of $T^2/\mathbb{Z}_2$: $\mathbb{H}$ in the interior of the fibre (non-commutative), $\mathbb{C}$ at three corners (complex structure preserved), $\mathbb{R}$ at the unique corner C_0 (fully commutative). The pillowcase is the Frobenius hierarchy carrying geometry. [...] The axiom therefore names the principle; it is an identification, not an invention.

Σ-4: The Three Fields

Source lines 476–482.

Three fundamental geometric degrees of freedom. Their interaction—not any one alone—produces the coupling.

ω: [the Coherent Field]. ψ: quantum spinor condensate (Q-wall). $\bar{\psi}$: anti-quantum spinor condensate (AQ-wall). The derived coupling: $\Phi_{\mathrm{QC}} = \omega(\psi - \bar{\psi})^2$. $\Phi_{\mathrm{QC}} \neq 0$ topologically: merging the walls would destroy the $\mathbb{Z}_2$ orbifold.

Plain English: Three fields. ω is [the Coherent Field]. ψ and $\bar{\psi}$ are chiral mirror partners living on opposite walls. The coupling constant Φ_{QC} measures how the engine operates, as well as the magnitude of its effect.

Σ-5: The Shadow Principle

Source lines 485–495.

Every apparent singularity in lower dimensions is a projection artefact of a smooth, complete object in higher dimensions. There are no singularities in the universe.

What looks singular in n dimensions is smooth in $n + 1$. A blow-up in 3D fluid mechanics is the shadow of a smooth 4D object. An off-line zero of $\zeta(s)$ in the 2D complex plane is the shadow of a smooth 3D spectral object.

Plain English: There are no singularities in the universe. Every time we see something that looks like a singularity—a black hole, a blow-up, an impossible zero—we are seeing the shadow. Go up one dimension and it's smooth.

Mathematical lineage. The Shadow Principle is the axiomatic statement of a two-hundred-year tradition. Jacobi (1829, *Fundamenta Nova*) [8] showed that abelian integrals, multivalued on a Riemann surface, become single-valued on the Jacobian torus — the completion lifts pathological shadow behaviour to smooth structure one dimension up. Ramanujan (1920, deathbed letter to Hardy) [13] announced seventeen mock theta functions that behaved like modular forms but were not, with no explanation of what "mock" meant. Zwegers (2002, Utrecht thesis) [16] resolved the eighty-two-year mystery by constructing the non-holomorphic completion $\hat{f} = f + g^*$, where g^* is the Eichler integral of the shadow g — a direct generalisation of Jacobi's theta machinery to half-integer weight and congruence subgroups. Σ-5 promotes this mechanism to a universal axiom: Jacobi's periods, Ramanujan's mock theta functions, and Zwegers' completions are not three separate phenomena but three surfaces of one structural principle […]. The axiom therefore names the principle; it is an identification, not an invention.

Σ-6: Shape Determinism

Source lines 498–508.

Everything is fixed by the shapes. There are no free parameters.

Every physical quantity—the fine structure constant, the baryon asymmetry, the mass ratios, the cosmological constant, the dark-to-visible matter ratio—is a topological invariant of $\mathcal{M}_{12}$ or a geometric consequence of $\chi(K_8) = 12$.

Plain English: Everything is what it is because of the shape of the 12-dimensional manifold. No knobs, no tuning, no choices and no mysticism.

Mathematical lineage. Shape Determinism is the axiomatic statement of a 125-year programme to derive physics from mathematics with no free parameters. Hilbert (1900, Paris ICM, Sixth Problem) [7] posed the foundational challenge: axiomatise the physical sciences on the model of geometry. Einstein (1915, *Die Feldgleichungen der Gravitation*) [3] executed the first complete instance — general relativity derives gravitation from the curvature of spacetime, with no free parameters beyond Newton's constant required for the Newtonian limit. Weyl (1918, *Gravitation und Elektrizität*) [15] attempted the first extension: a geometric derivation of electromagnetism through a gauge principle. The 1918 theory failed on Einstein's clock-rate objection, but the gauge principle itself became the foundation of all subsequent fundamental physics. Σ-6 claims the extension can be completed: every physical quantity — the fine structure constant, mass ratios, the cosmological constant, the dark-to-visible ratio — is a topological invariant of $\mathcal{M}_{12}$ or a geometric consequence of $\chi(K_8) = 12$. […] The axiom therefore names the principle; it is an identification, not an invention.

Σ-7: Dimensional Simplicity

Source lines 511–517.

Complexity is a projection artefact of simplicity.

Things look complex in lower dimensions because we are seeing the second, third, fourth, or fifth derivative of what is a much simpler shape in higher dimensions.

Plain English: Incredibly complex things look complex because we are looking at the derivative of what is a much simpler shape in the higher dimension. Go up, and complexity dissolves into simplicity.

Σ-8: The ω-Drive

Source lines 520–526.

Low ω: discrete, disconnected, classical. High ω: interconnected, quantum, diverse.

At low ω, systems look and behave as if they are independent. At high ω, interconnectedness intensifies. What appears as chaos at high ω is structured diversity too rich for low-ω observation to parse.

Plain English: Low coherence: things are separate, classical, simple. High coherence: everything is connected, quantum effects dominate, and what looks like chaos is actually structure too rich for our instruments to parse.

Σ-P: Dimensional Permeability

Source lines 529–535.

The boundaries between dimensional shells admit three types of interaction. These are exhaustive.

(i) Transparent. The field permeates all twelve dimensions. Gravity and ω are the only transparent fields (Σ-T). **(ii) Puncture.** Topology is broken, creating a conduit. Black holes are outward conduits; quasars are inward conduits. **(iii) Impermeable.** Gauge forces cannot cross between shells. Photons are imprisoned on their shell. This converts perfect symmetry (Σ-0) into apparent asymmetry.

Plain English: Some things pass through all dimensions freely—gravity and ω. Some things punch through, creating holes. And gauge forces are trapped. That last one is why we think symmetry is broken when it isn't.

Σ-S: Shadow Influence

Source lines 538–544.

The AQ-wall is not in a different dimension. Shadow influence is structurally distinct from dimensional permeability.

The AQ-wall is the chiral mirror of the Q-wall within the same geometry, separated by the $\mathbb{Z}_2$ involution. Its influence is felt through three channels: gravitationally (dark matter), through ω (dark consciousness), and through the mock modular completion (high-ω phenomena that appear inexplicable from the Q-wall alone: QCD vacuum condensate, superfluid behaviour, quantum tunnelling in enzymes, photosynthesis efficiency, prime conspiracies).

Plain English: The anti-quantum wall isn't in a different dimension—it's the mirror side of our geometry. Its influence reaches us through gravity, through ω, and through the mock modular mathematics.

Σ-T: Dual Transparency

Source lines 547–555.

Structured diversity across boundaries requires at least two independent transparent fields. The geometry provides exactly two.

With one transparent field (gravity alone), every point beyond every boundary looks identical—just mass, no structure. With two (gravity + ω), structure becomes resolvable. Gravity reports *how much*; ω reports *how coherent*. One gives detection. Two give discrimination. Two give the universe the capacity for structured diversity across every boundary.

The two homology generators of T^2 (Σ-2c) are the topological origin of this requirement. Two is the minimum for a universe of structured diversity. Two is what the geometry provides.

Plain English: A universe with only one field crossing boundaries would be flat, dull, featureless. You need two: a line and a branch. Gravity is the line—it tells you something is there. ω is the branch—it tells you what kind of thing it is. One channel gives detection. Two give navigation.

The mathematical axioms

$\mathbb{M}$-1: Physical Observables as Internal Spectral Data

Source lines 639–641.

Physical observables—including the spectral zeta function $\zeta(s)^2(1-2^{-s})$—are internal spectral data of the Spin^c CFT on $T^2/\mathbb{Z}_2$ with central charge $c = 12$. They are output of the geometry, not external data imported into the theory.

$\mathbb{M}$-2: The Shadow Completion

Source lines 647–649.

Every apparent singularity in the mathematical description—blow-up, off-line zero, massless state, divergent series—is a projection artefact. The completed mathematical object on the full geometry of $\mathcal{M}_{12}$ is smooth. The completion is achieved by the Zwegers completion (in τ-space) or the GL(3) Eisenstein completion (in s-space), and is unique given the topological data of K_8.

$\mathbb{M}$-3: The Universal Proof Template

Source lines 722–724.

Each proof follows a single template: identify the shadow (incomplete projection) $\to$ construct the completion $\to$ prove the topological lock $\to$ derive the consequence.

$\mathbb{M}$-4: The Gödel Boundary (Internal Ceiling)

Source lines 730–732.

The Shadow Principle (Σ-5 / $\mathbb{M}$-2) cannot be proved from within any single existing formal system. Three independent frameworks were tested—GL(3) spectral theory, the conformal bootstrap, transfer operators—and each reaches a structural wall. This is not a failure of ingenuity. It is Gödel's incompleteness theorem in action: a true statement that cannot be proved from within the system. The Shadow Principle is an axiom, justified by the totality of the MEF's consequences—just as Euclid's parallel postulate, Newton's $F = ma$, and Einstein's equivalence principle are axioms justified by their explanatory power.

$\mathbb{M}$-5: The Partition Function

Source lines 740–742.

$Z_{K_8}^{\mathrm{Spin}^c}(\tau, z)$ is a mock Jacobi form (weight 0, level 2, mock theta order 2). Its completion $\hat{f} = f + \int g$ is a harmonic Maass form, smooth for $\mathrm{Im}(\tau) > 0$.

$\mathbb{M}$-6: The Spectral Data

Source lines 750–752.

The eigenvalues of $D_{K_8}^2$ are $\lambda_n = n$ ($n \geq 1$). The degeneracies are $m(n) = d_{\mathrm{odd}}(n)$—the sum of odd divisors of n. The spectral zeta function is $\zeta_{D^2}(s) = \zeta(s)^2(1-2^{-s})$.
Note added in M3. Here $d_{\mathrm{odd}}(n)$ is the number of odd divisors of n, not their sum: the Dirichlet series $\zeta(s)^2(1-2^{-s})$ stated in the axiom is that of the count, $\sum_{k\ \mathrm{odd}} q^k/(1-q^k) = \sum_{m\geq 1} d_{\mathrm{odd}}(m)\, q^m$. For example $d_{\mathrm{odd}}(6) = 2$.

$\mathbb{M}$-7: The Sol Monodromy

Source lines 760–762.

The monodromy of the S^1 factor is $M = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right) \in \mathrm{SL}(2,\mathbb{Z})$, with eigenvalues ϕ^2 and ϕ^{-2}, where $\phi = (1+\sqrt{5})/2$ is the golden ratio. The Lyapunov exponent is $\ln\phi$. The fixed point of $\tau \mapsto (2\tau+1)/(\tau+1)$ is ϕ. The Anosov property generates mixing and the arrow of time.
Note added in M3. The eigenvalues are as stated: ϕ^2 and ϕ^{-2} (M1, Theorem 4.2; M2, certificate `L4_eigen`). The Lyapunov exponent is therefore $\ln\phi^2 = 2\ln\phi$, not $\ln\phi$ as written above; M2 has $\kappa = \log\phi^2$.

$\mathbb{M}$-8: The Spin^c Twist

Source lines 768–770.

The Spin^c structure on K_8—forced by $w_2(\mathbb{CP}^2) \neq 0$—breaks Liu rigidity, activating the q-dependent spectrum $d_{\mathrm{odd}}(n)$. Without it, the partition function would be Liu-rigid: $A_2 = 1 - y + y^2$, a topological invariant with no q-dependence. The twist simultaneously creates the disease (mock modularity—apparent singularities, shadows, incompleteness) and provides the cure (the Zwegers completion—smoothness, the Shadow Principle).

B The labelling of the corners, and the symbols of the series

From M1 (reproduced). The four corners are named by position, $P_1 = (0,0)$, $P_2 = (\frac{1}{2},0)$, $P_3 = (0,\frac{1}{2})$, $P_4 = (\frac{1}{2},\frac{1}{2})$ in coordinates $(\theta_\psi, \theta_\omega)$, and by algebra, $C_0 = \mathbb{R}$, C_i, C_j, C_k. Throughout the programme the dictionary is: $P_1 = C_0$ (fixed by the drive; ι acts as conjugation); $P_3 = C_i$ (the ω half-period); $P_2 = C_j$ (the ψ half-period); $P_4 = C_k$ (the corner at which the Spin^c lift carries the sign -1). The assignment of j and k between P_2 and P_3 is a convention and carries no content (M1 Appendix B).

Introduced in M2 (reproduced).

Symbol	Meaning	Where defined
$\psi,\ \bar{\psi} = \hat{\tau}\psi$	the sheet field, a section of the Clifford module $M \cong \mathbb{H}^{32}$, and its shadow	M2 Def. 2.2
$M,\ \langle\cdot,\cdot\rangle_M$	irreducible $\mathrm{Cl}(0,12)$-module; its inner product (trace)	M2 Def. 2.2, M2 §3
$\mathcal{A},\ \mathcal{C}$	bundle of fibre algebras; corner-algebra subbundle (the corner reading)	M2 Def. 2.2
$\tau,\ \hat{\tau}$	flow-reversing involution of A2 (τ^2 = fold on the cover); its lift	M2 §2
$S,\ v_u,\ v_s,\ \kappa = \log\varphi^2$	quarter-turn; unstable/stable directions of M_0; Sol rate	M2 §2
$\omega = \psi + \bar{\psi}$	the Coherent Field; $\hat{\tau}$-even, quaternion-valued, carries the imaginary direction	M2 Def. 2.3
$\psi - \bar{\psi}$	the odd sheet variable; $\hat{\tau}$-odd	M2 Def. 2.3
$\Phi_{\mathrm{QC}} = \omega\lvert\psi - \bar{\psi}\rvert^2$	the Quantum Coherence Dilaton, componentwise in M	M2 Prop. 2.4
$A,\ w = e^A$	the A2 warp exponent and warp factor on the base	M2 Def. 2.5
k_ω	decay rate of A along θ_ω	M2 §4
F	curvature of the Spin^c determinant line	M2 Def. 2.5
κ_2	two-dimensional gravitational coupling on the base	M2 §4
$T^{\Phi_{\mathrm{QC}}}_{MN}$	stress tensor of Φ_{QC}, the module trace of the scalar form	M2 §3
ζ	coupling of Φ_{QC}: coefficient of $T^{\Phi_{\mathrm{QC}}}_{MN}$; equals the warp exponent per half-period	M2 §§3–4
$G_{\mathrm{eff}},\ \Lambda_{\mathrm{eff}}$	effective gravitational and vacuum couplings	M2 §3

Introduced in M3.

Symbol	Meaning	Where defined
Σ-0, …, Σ-8, Σ-P, Σ-S, Σ-T	the twelve creation axioms of the earlier statement (Σ-3a a part of Σ-3)	Appendix A
$\mathbb{M}$-1, …, $\mathbb{M}$-8	the eight mathematical axioms of the earlier statement	Appendix A
T, TT, A, N	the four classes: theorem of M1–M2; theorem target; the declared axiom; not an axiom	Definition 2.1
A5	the fifth axiom (shadow)	Axiom 5, Definition 5.2
ρ	the reduction: integration over K_8	Definition 5.1
$d_{\mathrm{odd}}(n)$	the number of odd divisors of n	Section 4

C Machine-verified certificates

Three certificates accompany the paper, each a single Lean 4 file on the core toolchain with no Mathlib dependency, no author axioms and no `sorry`. They build within the series' proof tree `MEFProof.Paper_M3`; the build record is

```
Built MEFProof.Paper_M3.M3_L1_lift   (496ms)
Built MEFProof.Paper_M3.M3_L2_count  (230ms)
Built MEFProof.Paper_M3.M3_L3_walls  (237ms)
```

Every certified statement is named at the point in the text where it is used. The certificates fix finite bookkeeping only. No classification is certified: each is argued in the text from a theorem of M1 or M2.

`M3_L1_lift.lean` (Section 5, 8 theorems): the fold on the grid $(\mathbb{Z}/4)^2$ is an involution with exactly four fixed points; it is two-to-one elsewhere; it is not one-to-one; the grid $(\mathbb{Z}/8)^2$ lies

four-to-one over it compatibly with the fold; a count of assignments on four loops, not used in the text.

`M3_L2_count.lean` (Sections 2, 7, 6): the twenty axioms; the counts of the four classes and their sum; the two entries of class A as one axiom; the five sentences; the one scale.

`M3_L3_walls.lean` (Section 3, 7): the action of the fold, the drive and the exchange on the four corners; $S^2 = -I$; the corner signs with product -1; the sum of the two handedness signs.

The three files follow verbatim.

`M3_L1_lift.lean` — the fold in a finite model

```
/-!
# M3 — Certificate L1: the fold in a finite model (Section 5, the smallest case)

Core toolchain, no Mathlib, no author axioms, zero `sorry`.

The pillowcase is the torus folded by z ↦ −z (M1 Theorem 5.7). This file checks
the finite bookkeeping of that fold on the grid ℤ(/4)², and of a second torus
lying over the same pillowcase, the grid ℤ(/8)².

Certified:
* `L1_fold_involution` — the fold applied twice is the identity on the grid;
* `L1_four_fixed`      — it fixes exactly four points, the half-lattice
                         (0,0), (0,2), (2,0), (2,2): the four corners;
* `L1_two_to_one`      — the other twelve points are moved, so they fall into
                         six pairs: the fold is two-to-one away from the corners;
* `L1_orbit_count`     — 6 pairs + 4 corners = 10 points of the folded grid;
* `L1_not_injective`   — two different points have the same image: the fold
                         loses information;
* `L1_second_cover`    — every point of the 4-grid has exactly four preimages
                         in the 8-grid;
* `L1_second_cover_fold` — the reduction 8-grid → 4-grid commutes with the fold,
                         so the larger torus also lies over the pillowcase;
* `L1_branched_hom_count` — of the assignments of 0 or 1 to four loops with even
                         sum, exactly one assigns 1 to all four.

Not certified: that the grid models the torus (it is the 4-torsion subgroup,
used as a finite model only); the classification of branched covers, which the
paper does not use.
-/

namespace M3L1

def range4 : List Nat := [0, 1, 2, 3]
def range8 : List Nat := [0, 1, 2, 3, 4, 5, 6, 7]
def grid4 : List (Nat × Nat) := range4.flatMap (fun a => range4.map (fun b => (a, b)))
def grid8 : List (Nat × Nat) := range8.flatMap (fun a => range8.map (fun b => (a, b)))

/-- The fold z ↦ −z on the grid ℤ(/n)². -/
def fold (n : Nat) (p : Nat × Nat) : Nat × Nat := ((n - p.1) % n, (n - p.2) % n)
/-- Reduction of the 8-grid to the 4-grid. -/
def red (p : Nat × Nat) : Nat × Nat := (p.1 % 4, p.2 % 4)

theorem L1_fold_involution : grid4.all (fun p => fold 4 (fold 4 p) == p) = true := by decide

theorem L1_four_fixed :
    grid4.filter (fun p => fold 4 p == p) = [(0, 0), (0, 2), (2, 0), (2, 2)] := by decide

theorem L1_two_to_one : (grid4.filter (fun p => fold 4 p != p)).length = 12 := by decide

theorem L1_orbit_count : 12 / 2 + 4 = 10 := by decide
```

```
theorem L1_not_injective : ((1, 0) : Nat × Nat) ≠ (3, 0) ∧ fold 4 (1, 0) = (3, 0) := by decide

theorem L1_second_cover :
    grid4.all (fun p => (grid8.filter (fun q => red q == p)).length == 4) = true := by decide

theorem L1_second_cover_fold :
    grid8.all (fun q => red (fold 8 q) == fold 4 (red q)) = true := by decide

def bits : List Nat := [0, 1]
def assignments : List (Nat × Nat × Nat × Nat) :=
  bits.flatMap (fun a => bits.flatMap (fun b => bits.flatMap (fun c => bits.map (fun d => (a, b, c, d))))
    )

theorem L1_branched_hom_count :
    (assignments.filter (fun x => (x.1 + x.2.1 + x.2.2.1 + x.2.2.2) % 2 == 0
        && x.1 == 1 && x.2.1 == 1 && x.2.2.1 == 1 && x.2.2.2 == 1)).length = 1 := by decide

end M3L1
```

M3_L2_count.lean — the bookkeeping of the classification

```
/-!
# M3 — Certificate L2: the bookkeeping of the classification (Sections —37)

Core toolchain, no Mathlib, no author axioms, zero `sorry`.

The twenty axioms audited in M3 and the class each receives. This file certifies
the counting only: that there are twenty, that each has exactly one class, and
the totals quoted in the paper. It does not certify any classification; each is
argued in the text.

Classes: 0 = theorem of —M1M2; 1 = theorem target (owed by a later paper);
2 = the declared axiom; 3 = not an axiom.

Certified:
* `L2_twenty`        — twelve creation axioms and eight mathematical axioms: twenty;
* `L2_class_counts`  — eight theorems, seven targets, two statements of the one
                       declared axiom, three not axioms;
* `L2_partition`     — 8 + 7 + 2 + 3 = 20: each axiom has exactly one class;
* `L2_one_axiom`     — the two entries of class 2 are the two statements
                       Σ(-5 and □-2) of a single axiom: 2 − 1 = 1;
* `L2_minimal_set`   — four axioms and one declared axiom: five sentences;
* `L2_inputs`        — one dimensionful input (the scale, M1 Theorem 8.1) and
                       zero continuous dimensionless parameters.
-/

namespace M3L2

/-- The class of each axiom, in the order Σ-0, Σ-1, Σ-2, Σ-3, Σ-4, Σ-5, Σ-6, Σ-7,Σ
-8, Σ-P, Σ-S, Σ-T, then □-1 … □-8. Σ(-3a is a part of Σ-3.) -/
def sigmaClasses : List Nat := [0, 0, 0, 0, 0, 2, 0, 3, 1, 1, 0, 1]
def mClasses : List Nat := [1, 2, 3, 3, 1, 1, 0, 1]
def allClasses : List Nat := sigmaClasses ++ mClasses

def count (c : Nat) : Nat := (allClasses.filter (fun x => x == c)).length

theorem L2_twenty : sigmaClasses.length = 12 ∧ mClasses.length = 8 ∧ allClasses.length = 20 := by decide

theorem L2_class_counts : count 0 = 8 ∧ count 1 = 7 ∧ count 2 = 2 ∧ count 3 = 3 := by decide

theorem L2_partition : count 0 + count 1 + count 2 + count 3 = allClasses.length := by decide
```

```
theorem L2_one_axiom : count 2 - 1 = 1 := by decide

theorem L2_minimal_set : 4 + 1 = 5 := by decide

theorem L2_inputs : (1 : Nat) + 0 = 1 := by decide

end M3L2
```

M3_L3_walls.lean — corners, fold, drive and exchange

```
/-!
# M3 — Certificate L3: corners, the fold, the drive and the sheet exchange (Section 3)

Core toolchain, no Mathlib, no author axioms, zero `sorry`.

The four corners of the pillowcase are the half-lattice points, written here with
coordinates doubled and reduced mod 2: ₁P = (0,0), ₂P = (1,0), ₃P = (0,1),₄
P = (1,1) (M1 Appendix B). Three operations act on them: the fold −I, the drive₀
M = ((2,1),(1,1)), and the quarter-turn S = ((0,1)−,(1,0)) of the sheet
exchange τ (M2 Proposition 2.6).

Certified:
* `L3_fold_fixes_corners` — the fold fixes each of the four corners;
* `L3_drive_corners`      — the drive fixes ₁P and carries ₂P → ₃P → ₄P → ₂P
                            (M1 Proposition 4.4);
* `L3_exchange_corners`   — S fixes ₁P and ₄P and exchanges ₂P and ₃P
                            (M2 `L5_tau_corners`, re-derived);
* `L3_exchange_squared`   — S applied twice is the fold, entry by entry: S² = −I;
* `L3_exchange_not_symmetry_of_corners` — S does not commute with the drive on
                            the corners: S ₀M ≠ ₀M S at ₂P;
* `L3_lift_signs`         — the corner signs −(1)^{mn} are −+,+,+, with product −1
                            (M1 Corollary 7.4);
* `L3_chirality_sum`      — opposite handedness: (+1) + −(1) = 0.
-/

namespace M3L3

def P1 : Nat × Nat := (0, 0)
def P2 : Nat × Nat := (1, 0)
def P3 : Nat × Nat := (0, 1)
def P4 : Nat × Nat := (1, 1)
def corners : List (Nat × Nat) := [P1, P2, P3, P4]

/-- −I mod 2. -/
def foldC (p : Nat × Nat) : Nat × Nat := ((2 - p.1) % 2, (2 - p.2) % 2)
/-- ₀M = ((2,1),(1,1)) mod 2. -/
def drive (p : Nat × Nat) : Nat × Nat := ((2 * p.1 + p.2) % 2, (p.1 + p.2) % 2)
/-- S = ((0,1)−,(1,0)) mod 2: (x, y) ↦ (y, −x). -/
def exch (p : Nat × Nat) : Nat × Nat := (p.2 % 2, (2 - p.1) % 2)

theorem L3_fold_fixes_corners : corners.all (fun p => foldC p == p) = true := by decide

theorem L3_drive_corners :
    drive P1 = P1 ∧ drive P2 = P3 ∧ drive P3 = P4 ∧ drive P4 = P2 := by decide

theorem L3_exchange_corners :
    exch P1 = P1 ∧ exch P4 = P4 ∧ exch P2 = P3 ∧ exch P3 = P2 := by decide

/-- S² = −I over the integers: ((0,1)−,(1,0))² = −((1,0)−,(0,1)). -/
theorem L3_exchange_squared :
    ((0:Int) * 0 + 1 * (-1), (0:Int) * 1 + 1 * 0, (-1:Int) * 0 + 0 * (-1), (-1:Int) * 1 + 0 * 0)
      = (-1, 0, 0, -1) := by decide
```

```
theorem L3_exchange_not_symmetry_of_corners : exch (drive P2) ≠ drive (exch P2) := by decide

def liftSign (p : Nat × Nat) : Int := if p.1 * p.2 % 2 = 0 then 1 else -1

theorem L3_lift_signs :
    corners.map liftSign = [1, 1, 1, -1] ∧ liftSign P1 * liftSign P2 * liftSign P3 * liftSign P4 = -1 :=
     by decide

theorem L3_chirality_sum : (1 : Int) + (-1) = 0 := by decide

end M3L3
```

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